

Cambridge (CIE) IGCSE Maths
Paper 3: Calculator (Set F)
Core

Saturday 26 April 2025

Morning (Time: 1 hour 30 minutes)

Total marks

/ 81

Instructions

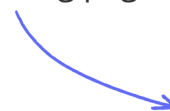
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- You should use a scientific calculator where appropriate.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

Materials

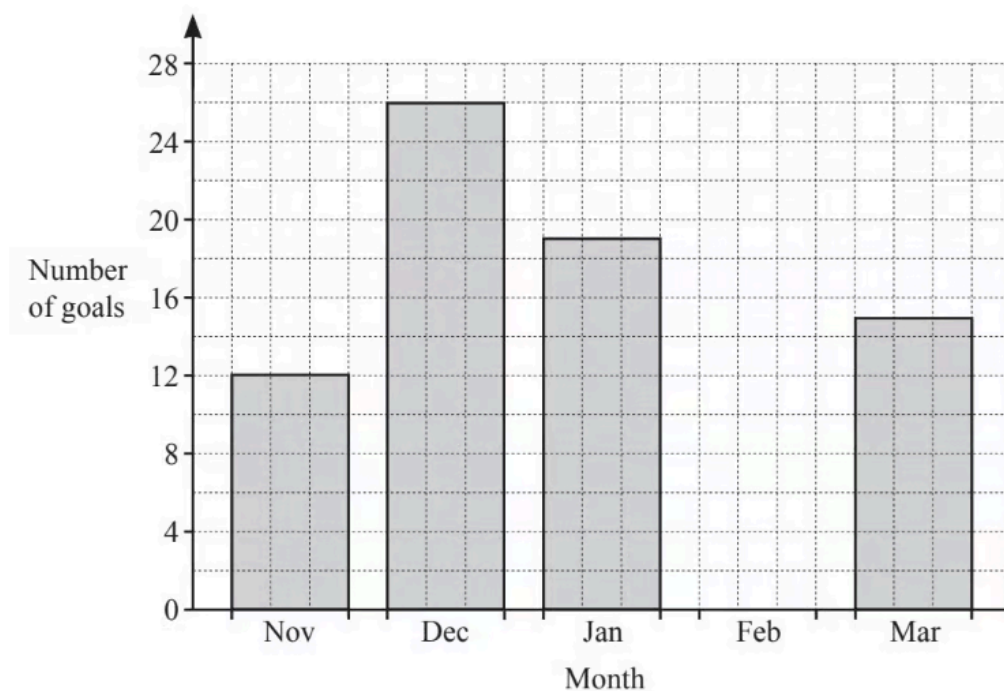
- [List of formulas](#)

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- 1 José manages a football team.
He records the number of goals scored by the team for each of five months.
Some of the results are shown on the bar chart.



- i) In February, 10 goals were scored.

Complete the bar chart.

[1]

- ii) Write down the month in which most goals were scored.

[1]

- iii) Find the total number of goals scored.

[1]

- iv) Calculate the mean number of goals scored each month.

[1]

(4 marks)

2 Write down all the factors of 15.

(2 marks)

3 (a) Edelgard tries to calculate $\frac{68 + 18}{9 - 5}$

She types into her calculator $68 + 18 \div 9 - 5$.

Explain why this does not give Edelgard the correct answer.

(1 mark)

(b) Work out the correct answer to $\frac{68 + 18}{9 - 5}$.

(1 mark)

4 Find the value of

i) $\sqrt{729}$

[1]

ii) 18^3 ,

[1]

iii) 7^0 .

[1]

(3 marks)

5 (a) Write down the temperature that is 7°C below -3°C .

..... $^{\circ}\text{C}$

(1 mark)

(b) Work out the difference in temperature between -4°C and 4°C .

..... $^{\circ}\text{C}$

(1 mark)

6 Sean is the manager of a museum.

He buys a Chinese pot costing 1200 yuan.

The exchange rate is $\$1 = 6.4$ yuan.

Work out the cost of this pot in dollars.

$\$$

(1 mark)

7 $W = 3a + 5c$

Find the value of W when $a = 6$ and $c = 2$.

$W =$

(2 marks)

8 In triangle ABC , $BC = 7.6\text{cm}$ and $AC = 6.2\text{cm}$.

Using a ruler and compasses only, construct triangle ABC .

Leave in your construction arcs.

The side AB has been drawn for you.



(2 marks)

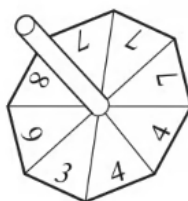
9 Calculate.

$$\frac{4.85 \times 6.14}{8.91 + 3.89}$$

Give your answer correct to 2 decimal places.

(2 marks)

10 (a) The diagram shows a fair 8-sided spinner.



The numbers on the spinner are 3, 4, 4, 7, 7, 7, 8 and 9.

The spinner is spun once.

Write down the probability that the spinner lands on

i) the number 7,

[1]

ii) a number greater than 2.

[1]

(2 marks)

(b) The spinner is spun 160 times.

Work out the expected number of times the spinner lands on the number 7.

.....

(1 mark)

- 11** Rachel has a pond in her garden in the shape of a circle.
The circumference of the pond is 4.25m.

Calculate the diameter of the pond.
Give your answer in centimetres.

..... cm

(3 marks)

- 12** Dina has a set of 12 cards. These are the numbers on the cards.

3	4	1	3	2	1	3	4	2	2	1	3
---	---	---	---	---	---	---	---	---	---	---	---

Work out

i) the median,

[2]

ii) the mode,

[1]

iii) the mean,

[2]

iv) the range.

[1]

(6 marks)

- 13 (a)** Louise leaves home at 09 55 and cycles the 5.6 km to the supermarket at a constant speed.
She takes 15 minutes to complete the journey.

Write down the time she arrives at the supermarket.

[1]

(1 mark)

- (b)** Calculate Louise's average speed from her home to the supermarket

i) in kilometres per hour,

.....km/h [1]

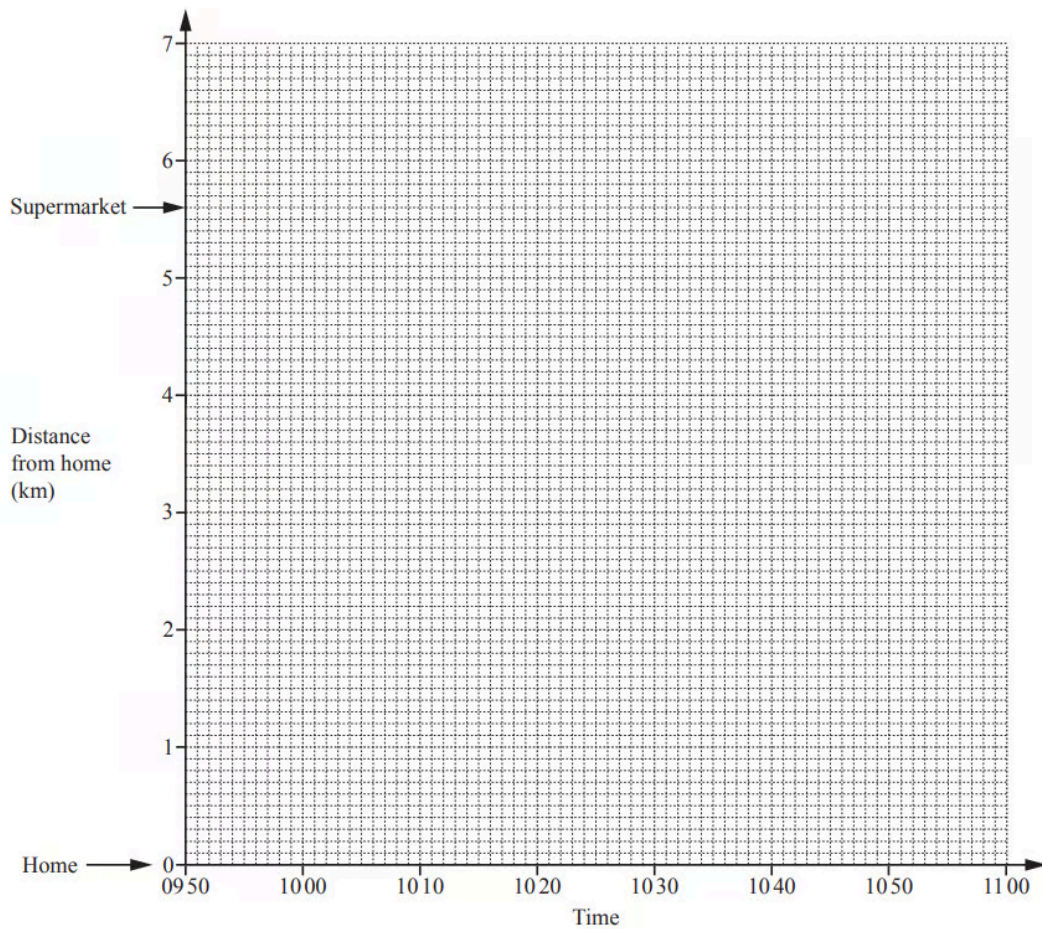
ii) in metres per second, giving your answer correct to 1 decimal place.

..... m/s [2]

(3 marks)

- (c)** Louise stays at the supermarket for 23 minutes.

On the grid, draw the travel graph of her journey from home and her stay at the supermarket.



(2 marks)

- (d) Louise's mother leaves home at 10 07 to meet Louise at the supermarket. She cycles at a constant speed of 28 km/h.

i) Work out how long she takes for the 5.6 km journey. Give your answer in minutes.

.....min [2]

ii) On the grid, show her mother's journey.

[1]

(3 marks)

(e) They cycle home together at a constant speed and arrive at 1054.

i) On the grid, show their journey home.

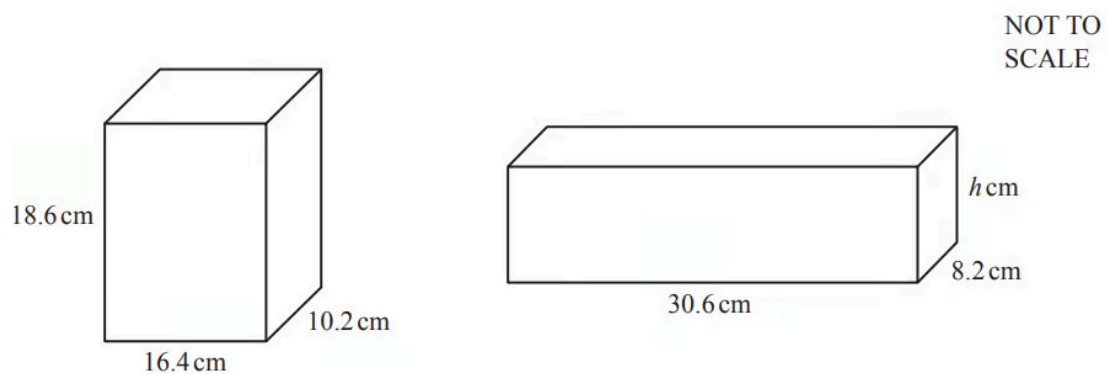
[1]

ii) Calculate, in km/h, their constant speed on the journey home.

.....km/h [2]

(3 marks)

14 (a) These two cuboids have the same volume.

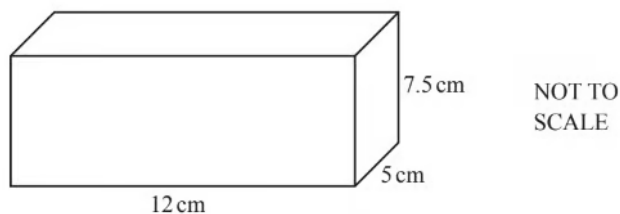


Find the value of h .

$h = \dots\dots\dots$

(3 marks)

(b)



Calculate the total surface area of the cuboid.

$\dots\dots\dots\text{cm}^2$

(3 marks)

- 15 The diagram shows a rectangle with length $7a$ and width $2a$.



Write an expression, in its simplest form, for

i) the perimeter,

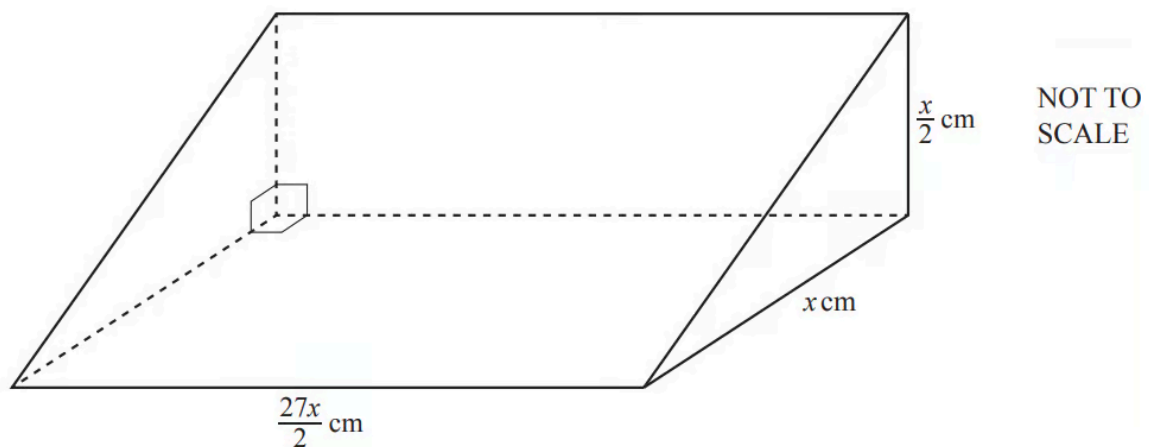
[2]

ii) the area.

[2]

(4 marks)

16



The diagram shows a prism with a right-angled triangle as its cross-section. The volume of the prism is 1000 cm^3 .

Calculate the value of x .

$x = \dots\dots\dots$

(4 marks)

- 17** A bag contains red marbles, green marbles and blue marbles only.
The ratio of the number of marbles of each colour is

red : green : blue = 12 : 5 : 2.

There are 112 more red marbles than green marbles.

Work out the number of blue marbles.

(2 marks)

- 18** \$8400 is invested for 2 years at a rate of 3.5% per year compound interest.
Work out the total amount of interest earned by the end of the 2 years.

\$.....

(3 marks)

19 i) Find the highest common factor (HCF) of 36 and 90.

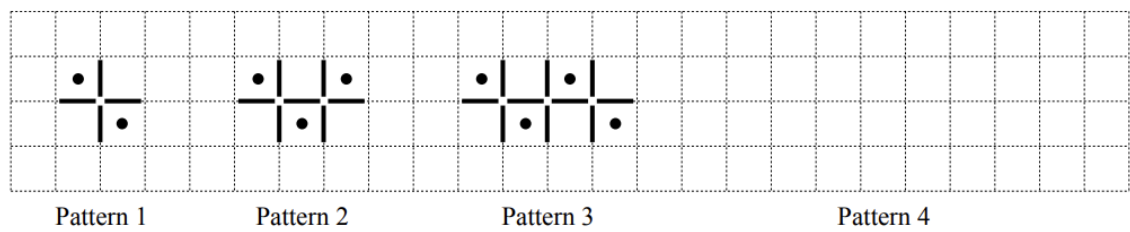
[2]

ii) Find the lowest common multiple (LCM) of 36 and 90.

[2]

(4 marks)

20 (a) A sequence of patterns is made using lines and dots. The first three patterns in the sequence are shown below.



Draw Pattern 4 on the grid.

(1 mark)

(b) Complete the table.

Pattern	1	2	3	4		10
Number of dots	2	3				
Number of lines	4	7				

(4 marks)

(c) Find an expression, in terms of n , for

i) the number of dots in Pattern n ,

[1]

ii) the number of lines in Pattern n .

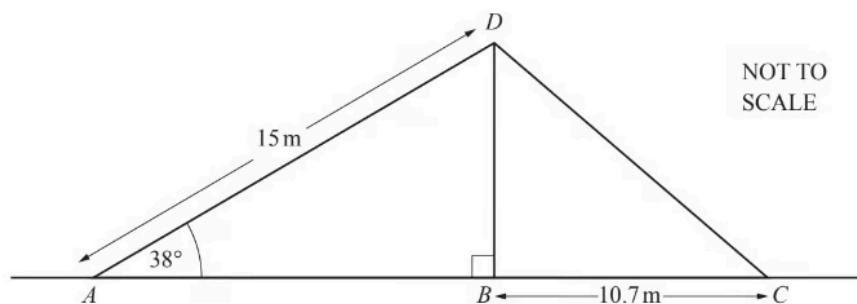
[2]

(3 marks)

(d) A pattern has 76 lines. Work out how many **dots** are in this pattern.

(2 marks)

21 (a)



A vertical flagpole, BD , stands on horizontal ground and is held by two ropes, AD and CD .

$AD = 15\text{m}$, $BC = 10.7\text{m}$ and angle $DAB = 38^\circ$.

Using trigonometry, calculate BD .

$BD = \dots\dots\dots\text{m}$

(2 marks)

(b) Calculate CD .

$CD = \dots\dots\dots\text{m}$

(2 marks)